```sh
sh kops.sh
```

```sh
export KOPS_STATE_STORE=s3://reyaz-kops-testbkt143.k8s.local
```

```sh
kops validate cluster --wait 10m
```

```sh
kubectl get nodes
```

```bash
cat <<EOF > selector.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lb-deployment
  labels:
    app: bank
spec:
  replicas: 2
  selector:
    matchLabels:
      app: bank
  template:
    metadata:
      labels:
        app: bank
    spec:
      nodeSelector:
        node-name: lb-node
      containers:
      - name: cont1
        image: reyadocker/internetbankingrepo:latest
EOF
```
```bash
vi nodeselector.yml

kubectl apply -f nodeselector.yml

kubectl get pods -o wide
```

```bash
kubectl get events
```

```bash
kubectl logs lb-deployment-7784c9bfb5-fv5qd

kubectl get events
```

```bash
kubectl describe pod lb-deployment-7784c9bfb5-fv5qd
```

```bash
kubectl get nodes

kubectl edit node i-004f9b5037e50bcfd
```

Edit node-name: ib-node like the following file:

Or use this command :

```bash
kubectl label node i-004f9b5037e50bcfd node-name=lb-node --overwrite
```

```
kubectl delete -f nodeselector. yml
Kibectl get pods
```

```
vi preferred.yml
```bash
cat <<EOF > preferred.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lb-deployment
  labels:
    app: bank
spec:
  replicas: 1
  selector:
```
```

```
    matchLabels:
      app: bank
  template:
    metadata:
      labels:
        app: bank
    spec:
      affinity:
        nodeAffinity:
          preferredDuringSchedulingIgnoredDuringExecution:
          - weight: 1
            preference:
              matchExpressions:
              - key: node-name
                operator: In
                values:
                - mb-node
      containers:
      - name: nginx
        image: nginx:1.14.2
        ports:
        - containerPort: 80
EOF
```
```
kubectl get nodes
```

```bash
kubectl apply -f preferred.yaml

kubectl get pods -o wide
```

```bash
kubectl delete -f preferred.yaml
```
```bash
cat <<EOF > required.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lb-deployment
  labels:
    app: bank
spec:
  replicas: 1
  selector:
    matchLabels:
```

```
      app: bank
  template:
    metadata:
      labels:
        app: bank
    spec:
      affinity:
        nodeAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            nodeSelectorTerms:
            - matchExpressions:
              - key: node-name
                operator: In
                values:
                - mb-node
      containers:
      - name: nginx
        image: nginx:1.14.2
        ports:
        - containerPort: 80
EOF
```
```bash
kubectl create -f required.yml

kubectl get pods -o wide

kubectl delete -f required.yml

clear
```

Why Master Node not schedule any job on itself the ans is tent keep as no -schedule check here by pasting masster id

```bash
kubectl get nodes

kubectl edit node i-02e9e62e21f186d3f
```

```
# List all nodes
[root@kops ~]# kubectl get nodes
NAME                  STATUS   ROLES          AGE   VERSION
i-004f9b5037e50bcfd   Ready    node           28m   v1.25.16
i-02e9e62e21f186d3f   Ready    control-plane   29m   v1.25.16
i-0e20e80d62aba4c2c   Ready    node           28m   v1.25.16
```

Try to edit node (cancelled)
[root@kops ~]# kubectl edit node i-02e9e62e21f186d3f
Edit cancelled, no changes made.

Taint GPU node to prevent scheduling regular pods
[root@kops ~]# kubectl taint nodes i-0e20e80d62aba4c2c hardware=gpu:NoSchedule
node/i-0e20e80d62aba4c2c tainted

```
```bash
vi taint.yml
```bash
cat <<EOF > taint.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ib-deployment
  labels:
    app: bank
spec:
  replicas: 1
  selector:
    matchLabels:
      app: bank
  template:
    metadata:
      labels:
        app: bank
    spec:
      containers:
      - name: cont1
        image: reyadocker/internetbankingrepo:latest
EOF
```

kubectl create -f taint.yml

kubectl get pods -o wide

kubectl delete -f taint.yml
```
```

```bash
vi taint.yml
```bash
cat <<EOF > tolerance.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ib-deployment
  labels:
    app: bank
spec:
  replicas: 1
  selector:
    matchLabels:
      app: bank
  template:
    metadata:
      labels:
        app: bank
    spec:
      containers:
      - name: cont1
        image: reyadocker/internetbankingrepo:latest
      tolerations:
      - key: "hardware"
        operator: "Equal"
        value: "gpu"
        effect: "NoSchedule"
EOF
```

kubectl create -f tolerance.yml

kubectl get pods -o wide

kubectl delete -f tolerance.yml

```

```bash
curl -fsSL -o get_helm.sh
https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3
chmod 700 get_helm.sh
./get_helm.sh
helm version
```

```
Install ARGOCD using HELM
---------------------------
helm repo add argo https://argoproj.github.io/argo-helm
helm repo update

kubectl create namespace argocd
helm install argocd argo/argo-cd --namespace argocd
kubectl get all -n argocd

EXPOSE ARGOCD SERVER:
--------------------
kubectl patch svc argocd-server -n argocd -p '{"spec":
{"type": "LoadBalancer"}}'

yum install jq -y

//export ARGOCD_SERVER='kubectl get svc argocd-server -n
argocd -o json | jq --raw-output
'.status.loadBalancer.ingress[0].hostname''

//echo $ARGOCD_SERVER

kubectl get svc argocd-server -n argocd -o json | jq
--raw-output .status.loadBalancer.ingress[0].hostname

The above command will provide load balancer URL to access
.GO CD


# Direct command — get and decode password in one line
kubectl -n argocd get secret argocd-initial-admin-secret -o
jsonpath="{.data.password}" | base64 -d

# Store password in a variable
export ARGO_PWD='kubectl -n argocd get secret
argocd-initial-admin-secret -o jsonpath="{.data.password}" |
base64 -d'

# Print the password
echo $ARGO_PWD

# The above command to provide password to access argo cd

TO GET ARGO CD PASSWORD:
```

```
------------------------
kubectl -n argocd get secret argocd-initial-admin-secret -o
jsonpath="{.data.password}" | base64 -d

export ARGO_PWD='kubectl -n argocd get secret
argocd-initial-admin-secret -o jsonpath="{.data.password}" |
base64 -d'

echo $ARGO_PWD

The above command to provide password to access argo cd
# Verify pods were created automatically by ArgoCD
kubectl get po   --> it created automatically from argocd

# Get services — copy paste the elb on browser
kubectl get svc  --> copy paste the elb on browser

Open ArgoCD load balancer

username: admin and password from above command

create app --> Application Name --> bankapp --> Project Name
--> default -->
Sync Policy --> Automatic --> Repository -->
https://github.com/ReyazShaik/ar-deploy.git
--> Path --> ./ --> CLuster URL --> NameSpace --> default

kubectl get po  --> it created automatically from argocd

kubectl get svc  --> copy paste the elb on browser

now modify replica as 5 in GitHub deploy.yml , automatically
argocd will deploy
```